

Ratchet: Literature-Only Transformer Optimization Survey

Ratchet project

2026-08-29

1 Scope

Ratchet preserves an authoritative full-transformer evaluator and prepares a future Intel-XPU research workflow. The present workspace has no qualified PyTorch XPU runtime. This paper therefore separates reviewed literature from project evidence and makes no project result claim.

2 Reviewed literature

This paper reports literature claims only, not measurements from this workspace. The reviewed primary-source set is [1, 3, 2, 5, 4, 6, 7, 8, 9]. The sources motivate future, hardware-gated investigation of IO-aware attention, compiler baselines, target-specific autotuning, and Intel XPU programming paths. They do not establish performance for this project or its unavailable runtime.

3 Catalogue summary

Catalogue fact	Value
Immutable experiment events	1
Reviewed primary sources	9

4 Evidence status

Verified event EVT-000001 / EXP-0001 records a no-run decision for ENV-0001: ENV-0001: PyTorch is not installed; empirical work is not permitted. The controller generated no candidate and recorded no compilation, correctness, timing, memory, profile, trace, counter, comparison, speedup, or current-best result. This is evidence of an unavailable execution environment, not a failed or slow implementation.

9 reviewed primary sources
Verified immutable archive projection
1 no-run event(s); 0 empirical event(s)
Literature synthesis only; no empirical conclusions

Figure 1: Evidence boundary for this literature-only build.

Evidence category	Count	Proportion
Reviewed primary sources	9	
No-run evidence events	1	
Empirical result events	0	

Figure 2: Evidence composition of the current build (bars are proportional to counts). Empirical result events remain zero until a hardware gate passes.

5 Next research direction

After FG-01 qualifies an Intel Arc/XPU runtime, test the queued protocol `PROTO-INTEL-0001@sha256:8497aaf0f9827ca6bedaea89d59a2146157324697fa918c0d84832ec6cdaa9c5:` After FG-01 qualifies Intel Arc/XPU, the full authoritative workload under `ratchet.intel.xpu.compiled.v1` may reduce steady-state latency versus `ratchet.intel.xpu.eager.v1` while preserving authoritative correctness and promotion gates. Its motivation is traceable to [1, 7, 8]. Until that hardware gate passes, this remains a protocol hypothesis rather than a project result.

References

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